

Line Integrals, Conservative Fields, and Surface Integrals

Lecture 11 for Multivariable Calculus

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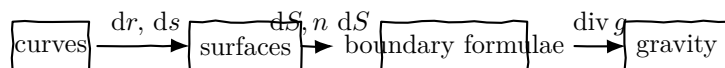
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Lecture Plan

Lecture 10 introduced line integrals, Green's theorem, conservative vector fields, and the first formulas for surface integrals. In Lecture 11 we slow down the last two topics.

1. Start from the familiar one-dimensional object: line integrals.
2. Add the special case where vector line integrals become easy: conservative fields.
3. Move from curves to surfaces by adding a second parameter.
4. Build scalar surface integrals first: density times surface area.
5. Add orientation and flux: normal component times surface area.
6. Compare the four integral types before introducing new theorems.
7. Discuss the Gauss formula and the Stokes formula as boundary-inside principles.
8. Apply Gauss' formula to the Newtonian gravity field.



Teaching Arc

The lecture grows in three layers:

geometry of integration $\longrightarrow$ orientation $\longrightarrow$ boundary equals inside.

Each new formula should answer one question: what object is being integrated, what direction has been chosen, and what derivative controls it?

Guiding Analogy

Line integrals have two common types:

$$\int_C f \, ds, \quad \int_C F \cdot dr.$$

Surface integrals have two parallel types:

$$\iint_S f \, dS, \quad \iint_S F \cdot n \, dS.$$

The scalar versions integrate density against geometric size. The vector versions integrate the component of a vector field in a chosen direction.

Unifying Principle

Many formulas in this lecture have the same structure:

integral over the boundary = integral of a derivative over the inside.

The derivative measures local change, local rotation, or local source strength. The boundary integral records the net effect seen from the outside. Orientation tells us which signs are positive.

1 Step 1: Review Line Integrals

1.1 Parametrized Curves

Let a curve C be parametrized by

$$r(t) = (x(t), y(t), z(t)), \quad a \leq t \leq b.$$

The velocity vector is

$$r'(t) = (x'(t), y'(t), z'(t)).$$

The small displacement vector is

$$dr = r'(t) dt,$$

and the small length element is

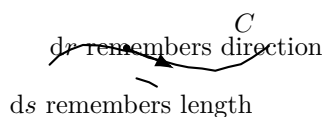
$$ds = \|r'(t)\| dt.$$

Two Small Objects

dr is a vector and remembers direction,

ds is a positive scalar length and forgets direction.

This is the source of the difference between the two types of line integrals.



1.2 Scalar Line Integrals

If f is a scalar function on C , then

$$\int_C f ds = \int_a^b f(r(t)) \|r'(t)\| dt.$$

If $f = 1$, then this is the length of C . If f is a density on a thin wire, then it is the mass of the wire.

Example. Let C be the line segment from $(0,0)$ to $(1,1)$, and let

$$f(x, y) = x + y.$$

Use

$$r(t) = (t, t), \quad 0 \leq t \leq 1.$$

Then

$$r'(t) = (1, 1), \quad \|r'(t)\| = \sqrt{2}.$$

Therefore

$$\int_C (x + y) ds = \int_0^1 2t\sqrt{2} dt = \sqrt{2}.$$

1.3 Vector Line Integrals

If

$$F = (P, Q, R)$$

is a vector field, then the vector line integral is

$$\int_C F \cdot dr = \int_a^b F(r(t)) \cdot r'(t) dt.$$

This integral measures the component of F in the direction of motion. If F is a force field, it computes work.

Tangential Component

Let

$$T(t) = \frac{r'(t)}{\|r'(t)\|}$$

be the unit tangent vector. Since

$$dr = T ds,$$

we have

$$\int_C F \cdot dr = \int_C (F \cdot T) ds.$$

Thus a vector line integral is a scalar line integral of the tangential component $F \cdot T$.

1.4 Orientation

If the direction of the curve is reversed, then ds stays the same but dr changes sign. Therefore

$$\int_{-C} f ds = \int_C f ds,$$

but

$$\int_{-C} F \cdot dr = - \int_C F \cdot dr.$$

Common Mistake

Do not replace dr by ds . They have different meanings:

$$dr = r'(t) dt, \quad ds = \|r'(t)\| dt.$$

The first is signed and directional. The second is unsigned length.

1.5 A Path-Dependence Example

Let

$$F(x, y) = (-y, x).$$

We compute the work from $(1, 0)$ to $(0, 1)$ along two paths.

Path 1: quarter circle. Use

$$r(t) = (\cos t, \sin t), \quad 0 \leq t \leq \frac{\pi}{2}.$$

Then

$$r'(t) = (-\sin t, \cos t),$$

and

$$F(r(t)) = (-\sin t, \cos t).$$

Thus

$$F(r(t)) \cdot r'(t) = 1,$$

so

$$\int_{C_1} F \cdot dr = \int_0^{\pi/2} 1 \, dt = \frac{\pi}{2}.$$

Path 2: straight line. Use

$$r(t) = (1 - t, t), \quad 0 \leq t \leq 1.$$

Then

$$r'(t) = (-1, 1), \quad F(r(t)) = (-t, 1 - t).$$

Therefore

$$F(r(t)) \cdot r'(t) = t + 1 - t = 1,$$

so

$$\int_{C_2} F \cdot dr = \int_0^1 1 \, dt = 1.$$

The two answers differ:

$$\frac{\pi}{2} \neq 1.$$

So this vector field is not path independent.

2 Step 2: Conservative Vector Fields

2.1 Definition and Main Property

A vector field F is called conservative if it is the gradient of a scalar potential function:

$$F = \nabla \phi.$$

In two dimensions,

$$F = (P, Q) = \left(\frac{\partial \phi}{\partial x}, \frac{\partial \phi}{\partial y} \right).$$

In three dimensions,

$$F = (P, Q, R) = \left(\frac{\partial \phi}{\partial x}, \frac{\partial \phi}{\partial y}, \frac{\partial \phi}{\partial z} \right).$$

Theorem 2.1 (Fundamental Theorem for Line Integrals). *If $F = \nabla \phi$ and C is any smooth curve from A to B , then*

$$\boxed{\int_C F \cdot dr = \phi(B) - \phi(A).}$$

Proof. Let C be parametrized by $r(t)$, $a \leq t \leq b$, with

$$r(a) = A, \quad r(b) = B.$$

Then by the chain rule,

$$\frac{d}{dt} \phi(r(t)) = \nabla \phi(r(t)) \cdot r'(t).$$

Therefore

$$\begin{aligned} \int_C F \cdot dr &= \int_a^b \nabla \phi(r(t)) \cdot r'(t) dt \\ &= \int_a^b \frac{d}{dt} \phi(r(t)) dt \\ &= \phi(r(b)) - \phi(r(a)) \\ &= \phi(B) - \phi(A). \end{aligned}$$

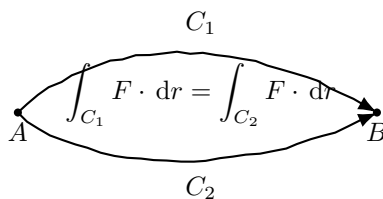
Meaning

For a conservative field, work depends only on the endpoints. The path does not matter:

$$\text{conservative field} \implies \text{path independence.}$$

Equivalently, every closed-loop integral is zero:

$$\oint_C F \cdot dr = 0.$$



in a conservative field, only endpoints matter

Equivalent Conditions for a Conservative Field

Let F be a smooth vector field on a connected region U . The following conditions are equivalent:

1. F is conservative; that is, there is a scalar potential ϕ such that

$$F = \nabla \phi.$$

2. Line integrals are path independent:

$$\int_{C_1} F \cdot dr = \int_{C_2} F \cdot dr$$

whenever C_1 and C_2 have the same initial point and final point.

3. Every closed-loop integral is zero:

$$\oint_C F \cdot dr = 0$$

for every closed curve C in U .

4. Fix a base point $A \in U$. The formula

$$\phi(B) = \int_A^B F \cdot dr$$

defines a well-defined potential function, independent of the path from A to B .

If U is simply connected, these are also equivalent to the curl-free condition:

$$\text{curl } F = 0.$$

In the plane, for $F = (P, Q)$, this condition is

$$\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 0.$$

Curl-Free Is Not Always Enough

The condition $\text{curl } F = 0$ is always necessary for a conservative field, but it is sufficient only when the domain has no holes, for example on a simply connected region.

2.2 How to Find a Potential in Two Dimensions

Suppose

$$F(x, y) = (P(x, y), Q(x, y)).$$

To find ϕ , solve

$$\phi_x = P, \quad \phi_y = Q.$$

Potential-Finding Method

1. Integrate P with respect to x :

$$\phi(x, y) = \int P(x, y) \, dx + h(y).$$

2. Differentiate this expression with respect to y .
3. Match the result with Q to determine $h'(y)$.
4. Integrate $h'(y)$ to find $h(y)$.

Example. Let

$$F(x, y) = (2xy + 3, x^2 + 2y).$$

We look for ϕ such that

$$\phi_x = 2xy + 3, \quad \phi_y = x^2 + 2y.$$

Integrate the first equation with respect to x :

$$\phi(x, y) = x^2y + 3x + h(y).$$

Then

$$\phi_y = x^2 + h'(y).$$

Matching with

$$Q = x^2 + 2y$$

gives

$$h'(y) = 2y.$$

Thus

$$h(y) = y^2 + C.$$

We can take

$$\boxed{\phi(x, y) = x^2y + 3x + y^2.}$$

Now any line integral can be computed from endpoints. For any curve from

$$A = (0, 1) \quad \text{to} \quad B = (2, -1),$$

we have

$$\int_C F \cdot dr = \phi(2, -1) - \phi(0, 1) = 3 - 1 = 2.$$

2.3 Curl Test in the Plane

For a vector field, curl measures local rotation. In Chinese, curl is usually called

旋度.

Definition of Curl

For a plane vector field

$$F(x, y) = (P(x, y), Q(x, y)),$$

the scalar curl is

$$\boxed{\text{curl } F = \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}.$$

It measures the tendency of the field to rotate around a point in the plane.

For a three-dimensional vector field

$$F(x, y, z) = (P(x, y, z), Q(x, y, z), R(x, y, z)),$$

the curl is the vector field

$$\boxed{\text{curl } F = (R_y - Q_z, P_z - R_x, Q_x - P_y).}$$

This is often remembered by the formal determinant

$$\boxed{\text{curl } F = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ P & Q & R \end{vmatrix}.$$

Expanding along the first row gives

$$\text{curl } F = (R_y - Q_z)\mathbf{i} - (R_x - P_z)\mathbf{j} + (Q_x - P_y)\mathbf{k},$$

which is the same as

$$(R_y - Q_z, P_z - R_x, Q_x - P_y).$$

This determinant is a mnemonic: the middle row contains differential operators, not ordinary numbers.

The direction of $\text{curl } F$ gives the local axis of rotation, and the magnitude measures the strength of rotation.

How the 3D Curl Reduces to the 2D Curl

A plane vector field

$$F(x, y) = (P(x, y), Q(x, y))$$

can be viewed as a three-dimensional vector field lying in the xy -plane:

$$\tilde{F}(x, y, z) = (P(x, y), Q(x, y), 0).$$

Here there is no z -component and no dependence on z . Using the three-dimensional curl formula,

$$\text{curl } \tilde{F} = (R_y - Q_z, P_z - R_x, Q_x - P_y),$$

with $R = 0$, $P_z = 0$, and $Q_z = 0$, we get

$$\text{curl } \tilde{F} = \left(0, 0, \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right).$$

Thus the two-dimensional scalar curl is the z -component of the three-dimensional curl:

$$\boxed{\text{curl}_{2D} F = \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = (\text{curl } \tilde{F}) \cdot \mathbf{k}.}$$

This is why positive scalar curl in the plane means counterclockwise local rotation, pointing in the positive z direction by the right-hand rule.

If

$$F = (P, Q) = \nabla \phi,$$

then

$$P = \phi_x, \quad Q = \phi_y.$$

Assuming second partial derivatives are continuous,

$$\frac{\partial Q}{\partial x} = \phi_{yx}, \quad \frac{\partial P}{\partial y} = \phi_{xy}.$$

Since mixed partial derivatives are equal,

$$\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 0.$$

Theorem 2.2 (Plane Conservative Field Test). *Let*

$$F = (P, Q)$$

be a vector field on a simply connected plane region, with continuous first partial derivatives. Then

$$F \text{ is conservative} \iff \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 0.$$

Role of Green's Theorem

If the scalar curl is zero, then Green's theorem gives

$$\oint_C P \, dx + Q \, dy = \iint_D 0 \, dA = 0$$

for every simple closed curve C bounding a region D inside the domain. Therefore closed-loop work is zero, so line integrals are path independent.

Thinking Question: What If the Domain Has a Hole?

The simply connected condition means the domain has no holes. What can go wrong if the domain does have a hole?

Consider the vector field

$$F(x, y) = \left(\frac{-y}{x^2 + y^2}, \frac{x}{x^2 + y^2} \right)$$

on the punctured plane

$$U = \mathbb{R}^2 \setminus \{(0, 0)\}.$$

One can check that

$$\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 0 \quad \text{for every } (x, y) \in U.$$

So locally the field looks curl-free.

Now take the unit circle

$$C : r(t) = (\cos t, \sin t), \quad 0 \leq t \leq 2\pi.$$

Then

$$F(r(t)) = (-\sin t, \cos t) = r'(t),$$

so

$$\oint_C F \cdot dr = \int_0^{2\pi} 1 \, dt = 2\pi.$$

Question for discussion:

If the curl is zero everywhere in U , why is the closed integral not zero?

What part of the Green's theorem argument fails? Notice that the unit circle surrounds the missing point, so it does not bound a disk contained completely inside U .

2.4 Three-Dimensional Version

For

$$F = (P, Q, R),$$

the curl is the vector field

$$\operatorname{curl} F = (R_y - Q_z, P_z - R_x, Q_x - P_y).$$

If

$$F = \nabla \phi,$$

then

$$\operatorname{curl} F = 0.$$

On a simply connected three-dimensional region with no holes, the converse is also true under the usual smoothness assumptions:

$$\operatorname{curl} F = 0 \implies F \text{ is conservative.}$$

Example in Three Dimensions

Let

$$F(x, y, z) = (y + z, x + z, x + y).$$

Notice that

$$\phi(x, y, z) = xy + xz + yz$$

satisfies

$$\nabla \phi = (y + z, x + z, x + y) = F.$$

Therefore for any curve from

$$A = (0, 0, 0) \quad \text{to} \quad B = (1, 2, 3),$$

$$\int_C F \cdot dr = \phi(B) - \phi(A) = 1 \cdot 2 + 1 \cdot 3 + 2 \cdot 3 = 11.$$

3 Step 3: Parametrized Surfaces

3.1 From Curves to Surfaces

A curve uses one parameter:

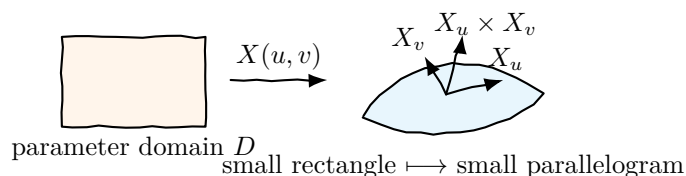
$$r(t).$$

A surface uses two parameters:

$$X(u, v) = (x(u, v), y(u, v), z(u, v)), \quad (u, v) \in D.$$

Here D is a region in the uv -plane.

Object	Parametrization	Small size
Curve	$r(t)$	$ds = \ r'(t)\ \, dt$
Surface	$X(u, v)$	$dS = \ X_u \times X_v\ \, du \, dv$



3.2 Tangent Vectors and Area Element

The tangent vectors are

$$X_u = \frac{\partial X}{\partial u}, \quad X_v = \frac{\partial X}{\partial v}.$$

For a small rectangle in the parameter domain with side lengths Δu and Δv , its image on the surface is approximately a parallelogram with side vectors

$$X_u \Delta u, \quad X_v \Delta v.$$

The area of this parallelogram is

$$\|X_u \times X_v\| \, \Delta u \, \Delta v.$$

Therefore

$$dS = \|X_u \times X_v\| \, du \, dv.$$

Vector Area Element

The vector

$$X_u \times X_v$$

is perpendicular to the surface. Its length gives the area scale, and its direction gives an orientation. Thus

$$\underbrace{\|X_u \times X_v\| \, du \, dv}_{\text{area element}}$$

and

$$\underbrace{(X_u \times X_v) \, du \, dv}_{\text{oriented vector area element}}$$

are closely related but not the same object.

3.3 Standard Parametrizations

Graph surface. If

$$S : z = g(x, y), \quad (x, y) \in D,$$

then

$$X(x, y) = (x, y, g(x, y)).$$

The tangent vectors are

$$X_x = (1, 0, g_x), \quad X_y = (0, 1, g_y).$$

Thus

$$X_x \times X_y = (-g_x, -g_y, 1),$$

and

$$\boxed{dS = \sqrt{1 + g_x^2 + g_y^2} \, dx \, dy.}$$

Sphere of radius R . Use

$$X(\phi, \theta) = (R \sin \phi \cos \theta, R \sin \phi \sin \theta, R \cos \phi),$$

where

$$0 \leq \phi \leq \pi, \quad 0 \leq \theta \leq 2\pi.$$

Then

$$dS = R^2 \sin \phi \, d\phi \, d\theta.$$

Cylinder of radius a . For the side surface of a cylinder,

$$X(\theta, z) = (a \cos \theta, a \sin \theta, z),$$

where

$$0 \leq \theta \leq 2\pi, \quad c \leq z \leq d.$$

Here

$$X_\theta = (-a \sin \theta, a \cos \theta, 0), \quad X_z = (0, 0, 1),$$

so

$$X_\theta \times X_z = (a \cos \theta, a \sin \theta, 0),$$

and

$$dS = a \, d\theta \, dz.$$

4 Step 4: Scalar Surface Integrals

4.1 Definition

If f is a scalar function on a surface S , then

$$\iint_S f \, dS = \iint_D f(X(u, v)) \|X_u \times X_v\| \, du \, dv.$$

If $f = 1$, this gives surface area:

$$\text{Area}(S) = \iint_S 1 \, dS.$$

If f is surface density, then the integral gives mass of a thin sheet:

$$M = \iint_S f \, dS.$$

Scalar Surface Integral

$$\iint_S f \, dS$$

adds up

surface density $\times$ surface area.

It does not depend on the orientation of the surface.

4.2 Graph Formula

For

$$S : z = g(x, y), \quad (x, y) \in D,$$

we have

$$\iint_S f \, dS = \iint_D f(x, y, g(x, y)) \sqrt{1 + g_x^2 + g_y^2} \, dx \, dy.$$

Example: Surface area of a paraboloid patch. Let

$$S : z = x^2 + y^2, \quad x^2 + y^2 \leq 1.$$

Then

$$g_x = 2x, \quad g_y = 2y,$$

so

$$dS = \sqrt{1 + 4x^2 + 4y^2} \, dx \, dy.$$

The area is

$$\text{Area}(S) = \iint_{x^2+y^2 \leq 1} \sqrt{1 + 4x^2 + 4y^2} \, dx \, dy.$$

Use polar coordinates:

$$\text{Area}(S) = \int_0^{2\pi} \int_0^1 \sqrt{1 + 4r^2} \, r \, dr \, d\theta.$$

Let

$$u = 1 + 4r^2, \quad du = 8r \, dr.$$

Then

$$\begin{aligned} \text{Area}(S) &= 2\pi \int_0^1 r \sqrt{1 + 4r^2} \, dr \\ &= 2\pi \cdot \frac{1}{8} \int_1^5 u^{1/2} \, du \\ &= \frac{\pi}{4} \cdot \frac{2}{3} u^{3/2} \Big|_1^5 \\ &= \boxed{\frac{\pi}{6} (5^{3/2} - 1)}. \end{aligned}$$

4.3 Example on a Sphere

Let S be the upper hemisphere

$$x^2 + y^2 + z^2 = R^2, \quad z \geq 0.$$

Compute

$$\iint_S z \, dS.$$

Use spherical coordinates on the surface:

$$X(\phi, \theta) = (R \sin \phi \cos \theta, R \sin \phi \sin \theta, R \cos \phi),$$

where

$$0 \leq \phi \leq \frac{\pi}{2}, \quad 0 \leq \theta \leq 2\pi.$$

Then

$$z = R \cos \phi, \quad dS = R^2 \sin \phi \, d\phi \, d\theta.$$

Therefore

$$\begin{aligned} \iint_S z \, dS &= \int_0^{2\pi} \int_0^{\pi/2} R \cos \phi \, R^2 \sin \phi \, d\phi \, d\theta \\ &= 2\pi R^3 \int_0^{\pi/2} \sin \phi \cos \phi \, d\phi \\ &= 2\pi R^3 \cdot \frac{1}{2} \\ &= \boxed{\pi R^3}. \end{aligned}$$

Scalar Surface Integral Workflow

1. Parametrize the surface $X(u, v)$, or write it as a graph.
2. Compute X_u , X_v , and $\|X_u \times X_v\|$.
3. Substitute the surface parametrization into f .
4. Integrate over the parameter domain.

5 Step 5: Oriented Surfaces and Flux

5.1 Orientation

For a surface, orientation means choosing one of the two normal directions. If n is the chosen unit normal vector, then $-n$ is the opposite orientation.

Orientable Surface

A surface is orientable if one can choose a continuous unit normal vector over the whole surface. For standard surfaces in this course, such as graphs, spheres, cylinders, and pieces of these surfaces, this is usually not a problem.

For a parametrized surface

$$X(u, v),$$

the cross product

$$X_u \times X_v$$

chooses one orientation. Swapping the parameter order gives

$$X_v \times X_u = -(X_u \times X_v),$$

which reverses the orientation.

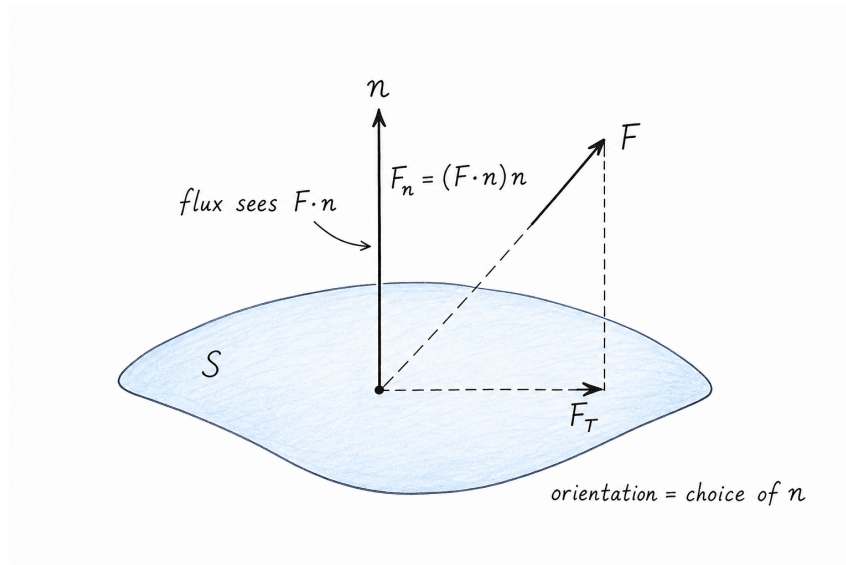


图 1: Essential picture of an oriented surface and flux. An orientation is a choice of unit normal n . Flux measures the normal component $F \cdot n$; the tangential component of F does not cross the surface, so it does not contribute to flux.

5.2 Definition of Flux

Let F be a vector field and let S be an oriented surface with unit normal n . The vector surface integral, also called flux, is

$$\iint_S F \cdot n \, dS.$$

It measures how much of the vector field passes through the surface in the chosen normal direction.

Using a parametrization, we can combine $n \, dS$ into one vector area element:

$$n \, dS = (X_u \times X_v) \, du \, dv$$

when the parametrization has the chosen orientation. Therefore

$$\iint_S F \cdot n \, dS = \iint_D F(X(u, v)) \cdot (X_u \times X_v) \, du \, dv.$$

Orientation Changes the Sign

The scalar surface integral

$$\iint_S f \, dS$$

does not change when orientation is reversed. But flux changes sign:

$$\iint_{-S} F \cdot n \, dS = - \iint_S F \cdot n \, dS.$$

5.3 Graph Formula for Flux

Let

$$S : z = g(x, y), \quad (x, y) \in D.$$

Use the parametrization

$$X(x, y) = (x, y, g(x, y)).$$

Then

$$X_x \times X_y = (-g_x, -g_y, 1),$$

which points upward because its third component is positive. Therefore, for upward orientation,

$$\boxed{\iint_S F \cdot n \, dS = \iint_D F(x, y, g(x, y)) \cdot (-g_x, -g_y, 1) \, dx \, dy.}$$

For downward orientation, use the negative vector:

$$(g_x, g_y, -1).$$

Graph Orientations

$$z = g(x, y) : \quad \text{upward vector area} = (-g_x, -g_y, 1) \, dx \, dy.$$

$$x = h(y, z) : \quad \text{positive } x\text{-direction vector area} = (1, -h_y, -h_z) \, dy \, dz.$$

$$y = k(x, z) : \quad \text{positive } y\text{-direction vector area} = (-k_x, 1, -k_z) \, dx \, dz.$$

5.4 Flux Example: Graph Surface

Let

$$S : z = x^2 + y^2, \quad x^2 + y^2 \leq 1,$$

with upward orientation. Let

$$F(x, y, z) = (0, 0, z).$$

Compute the flux

$$\iint_S F \cdot n \, dS.$$

Since

$$g(x, y) = x^2 + y^2,$$

we have

$$g_x = 2x, \quad g_y = 2y.$$

The upward vector area element is

$$(-2x, -2y, 1) \, dx \, dy.$$

On the surface,

$$F(x, y, g(x, y)) = (0, 0, x^2 + y^2).$$

Thus

$$F(x, y, g(x, y)) \cdot (-2x, -2y, 1) = x^2 + y^2.$$

Therefore

$$\begin{aligned} \iint_S F \cdot n \, dS &= \iint_{x^2+y^2 \leq 1} (x^2 + y^2) \, dx \, dy \\ &= \int_0^{2\pi} \int_0^1 r^2 \, r \, dr \, d\theta \\ &= 2\pi \cdot \frac{1}{4} \\ &= \boxed{\frac{\pi}{2}}. \end{aligned}$$

If the downward orientation were chosen, the answer would be

$$-\frac{\pi}{2}.$$

5.5 Flux Example: Cylinder

Let S be the side surface of the cylinder

$$x^2 + y^2 = a^2, \quad 0 \leq z \leq h,$$

with outward orientation. Let

$$F(x, y, z) = (x, y, 0).$$

Parametrize by

$$X(\theta, z) = (a \cos \theta, a \sin \theta, z), \quad 0 \leq \theta \leq 2\pi, \quad 0 \leq z \leq h.$$

Then

$$X_\theta = (-a \sin \theta, a \cos \theta, 0), \quad X_z = (0, 0, 1),$$

and

$$X_\theta \times X_z = (a \cos \theta, a \sin \theta, 0),$$

which points outward. On the surface,

$$F(X(\theta, z)) = (a \cos \theta, a \sin \theta, 0).$$

Hence

$$F(X) \cdot (X_\theta \times X_z) = a^2 \cos^2 \theta + a^2 \sin^2 \theta = a^2.$$

Thus

$$\iint_S F \cdot n \, dS = \int_0^h \int_0^{2\pi} a^2 \, d\theta \, dz = \boxed{2\pi a^2 h}.$$

5.6 Flux Example: Sphere

Let S be the sphere

$$x^2 + y^2 + z^2 = R^2$$

with outward orientation, and let

$$F(x, y, z) = (x, y, z).$$

On the sphere, the outward unit normal is

$$n = \frac{1}{R}(x, y, z).$$

Therefore

$$F \cdot n = (x, y, z) \cdot \frac{1}{R}(x, y, z) = \frac{x^2 + y^2 + z^2}{R} = R.$$

Thus

$$\iint_S F \cdot n \, dS = \iint_S R \, dS = R \text{Area}(S) = \boxed{4\pi R^3}.$$

Flux Interpretation

If F is a fluid velocity field, then

$$\iint_S F \cdot n \, dS$$

is the net rate at which fluid crosses the surface in the chosen normal direction. If $F \cdot n > 0$, flow goes with the chosen normal. If $F \cdot n < 0$, flow goes against it.

6 Step 6: Four Integrals Side by Side

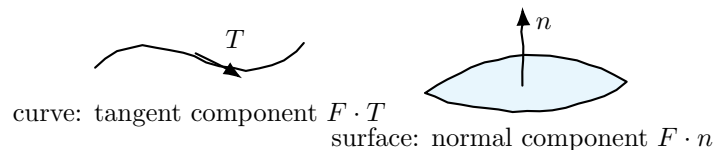
Object	Scalar type	Vector type	Direction issue
Curve C	$\int_C f \, ds$	$\int_C F \cdot dr$	orientation of path
Surface S	$\iint_S f \, dS$	$\iint_S F \cdot n \, dS$	orientation of normal

Unified Pattern

$$\text{scalar type} = \int \text{density} \times \text{geometric size}.$$

$$\text{vector type} = \int \text{chosen-direction component} \times \text{geometric size}.$$

For curves, the chosen direction is tangent to the path. For surfaces, the chosen direction is normal to the surface.



6.1 Common Computation Patterns

Task	Formula	Notes
Curve length	$\int_C 1 \, ds$	scalar line integral
Work	$\int_C F \cdot dr$	vector line integral
Surface area	$\iint_S 1 \, dS$	scalar surface integral
Mass of a sheet	$\iint_S \rho \, dS$	ρ is surface density
Flux through a surface	$\iint_S F \cdot n \, dS$	orientation matters

Formula Choice

Before computing, identify the type of integral.

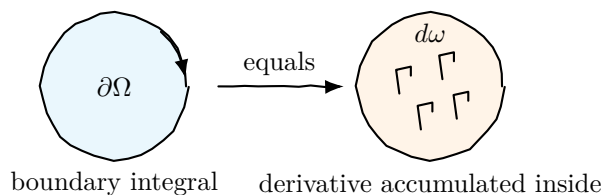
ds or dr ?

dS or $n \, dS$?

Most errors in this topic come from choosing the wrong geometric element.

7 Step 7: Gauss Formula and Stokes Formula

The formulas in this section are global versions of the same idea: an integral over a boundary is controlled by what happens inside.



7.1 Divergence and the Gauss Formula

For a three-dimensional vector field

$$F(x, y, z) = (P(x, y, z), Q(x, y, z), R(x, y, z)),$$

the divergence, called

散度

in Chinese, is

$$\operatorname{div} F = \nabla \cdot F = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z}.$$

It measures the local source strength of the vector field. Positive divergence means net flow is created near the point; negative divergence means net flow is absorbed.

Theorem 7.1 (Gauss Formula, or Divergence Theorem). *Let E be a solid region in $\mathbb{R}^3$ with closed boundary surface ∂E , oriented by the outward unit normal n . If F is smooth, then*

$$\iint_{\partial E} F \cdot n \, dS = \iiint_E \operatorname{div} F \, dV.$$

Meaning of Gauss Formula

The left side is total outward flux through the boundary. The right side is total source strength inside the solid.

$$\text{outward flux through boundary} = \text{total divergence inside.}$$

The surface must be closed, and the normal must point outward.

Example. Let E be the ball

$$x^2 + y^2 + z^2 \leq R^2,$$

and let

$$F(x, y, z) = (x, y, z).$$

The boundary ∂E is the sphere of radius R , oriented outward. Since

$$\operatorname{div} F = 1 + 1 + 1 = 3,$$

Gauss' formula gives

$$\iint_{\partial E} F \cdot n \, dS = \iiint_E 3 \, dV = 3 \cdot \frac{4}{3} \pi R^3 = \boxed{4\pi R^3}.$$

This matches the direct sphere computation from the previous section.

7.2 Stokes Formula

Stokes' formula connects a line integral around a boundary curve to the flux of curl through a surface.

Theorem 7.2 (Stokes Formula). *Let S be an oriented surface with unit normal n , and let ∂S be its boundary curve with the induced positive orientation. If F is smooth, then*

$$\oint_{\partial S} F \cdot dr = \iint_S (\text{curl } F) \cdot n \, dS.$$

Boundary Orientation

The orientation of ∂S is determined by the right-hand rule. If the thumb points along the chosen normal n , then the fingers curl in the positive direction of the boundary curve.

Meaning of Stokes Formula

The left side is circulation around the boundary. The right side is the flux of curl through the surface:

$$\text{circulation around boundary} = \text{total rotation through the surface.}$$

The surface itself can bend, but the boundary curve and the orientation control the sign.

Example. Let

$$F(x, y, z) = \left(-\frac{y}{2}, \frac{x}{2}, 0 \right),$$

and let S be the unit disk in the plane $z = 0$, oriented upward. Its boundary ∂S is the unit circle oriented counterclockwise when viewed from above.

The curl is

$$\text{curl } F = (0, 0, 1).$$

Therefore Stokes' formula gives

$$\oint_{\partial S} F \cdot dr = \iint_S (0, 0, 1) \cdot (0, 0, 1) \, dS = \iint_S 1 \, dS = \boxed{\pi}.$$

So this vector field is normalized so that circulation around a planar curve measures the area it encloses.

Stokes Formula Contains Green's Theorem

Take a plane vector field

$$F(x, y) = (P(x, y), Q(x, y))$$

and view it in space as

$$\tilde{F}(x, y, z) = (P(x, y), Q(x, y), 0).$$

For a flat region D in the xy -plane with upward normal $\mathbf{k}$,

$$(\text{curl } \tilde{F}) \cdot \mathbf{k} = Q_x - P_y.$$

Thus Stokes' formula becomes

$$\oint_{\partial D} P \, dx + Q \, dy = \iint_D (Q_x - P_y) \, dA,$$

which is exactly Green's theorem.

7.3 The Unified Principle

The fundamental theorem of calculus, Green's theorem, Stokes' theorem, and Gauss' theorem are not separate miracles. They are different dimensions of the same principle:

$$\boxed{\int_{\partial\Omega} \omega = \int_{\Omega} d\omega.}$$

This compact formula is called the generalized Stokes theorem. In this notation, Ω is a region, $\partial\Omega$ is its boundary, ω is the object being integrated, and $d\omega$ is its derivative.

Boundary Equals Accumulated Interior Derivative

net effect on the boundary = sum of local changes inside.

The boundary may be two endpoints, a closed curve, or a closed surface. The derivative may be an ordinary derivative, a scalar curl, a vector curl, or a divergence.

Formula	Region Ω	Boundary $\partial\Omega$	Interior derivative
Fundamental theorem	interval	two endpoints	ϕ'
Gradient theorem	curve C	endpoints A, B	$\nabla\phi$ along the curve
Green theorem	plane region D	closed curve ∂D	$Q_x - P_y$
Stokes theorem	surface S	boundary curve ∂S	$\text{curl } F$
Gauss theorem	solid E	closed surface ∂E	$\text{div } F$

Applying the General Formula

The symbol $\wedge$ is a bookkeeping device for oriented area and volume elements. If this notation is new, focus on the last equality in each case.

1. Fundamental theorem of calculus. Take

$$\Omega = [a, b], \quad \partial\Omega = \{b\} - \{a\}, \quad \omega = \phi.$$

Then

$$d\omega = \phi'(x) \, dx.$$

The general formula gives

$$\int_{\partial[a,b]} \phi = \int_a^b \phi'(x) \, dx,$$

that is,

$$\phi(b) - \phi(a) = \int_a^b \phi'(x) \, dx.$$

2. Fundamental theorem for line integrals. Take $\Omega = C$, an oriented curve from A to B , and take

$$\omega = \phi.$$

Then

$$d\omega = d\phi = \nabla\phi \cdot dr.$$

Since

$$\partial C = \{B\} - \{A\},$$

the general formula gives

$$\int_{\partial C} \phi = \int_C d\phi,$$

so

$$\phi(B) - \phi(A) = \int_C \nabla\phi \cdot dr.$$

Thus if $F = \nabla\phi$, then

$$\int_C F \cdot dr = \phi(B) - \phi(A).$$

3. Green's theorem. Take $\Omega = D$, a plane region, and take

$$\omega = P \, dx + Q \, dy.$$

Then

$$d\omega = (Q_x - P_y) \, dx \wedge dy.$$

The general formula gives

$$\int_{\partial D} \omega = \int_D d\omega,$$

which becomes

$$\oint_{\partial D} P \, dx + Q \, dy = \iint_D (Q_x - P_y) \, dA.$$

4. Stokes' theorem. Take $\Omega = S$, an oriented surface, and take

$$\omega = P \, dx + Q \, dy + R \, dz = F \cdot dr.$$

Then

$$\begin{aligned} d\omega &= (R_y - Q_z) \, dy \wedge dz + (P_z - R_x) \, dz \wedge dx \\ &\quad + (Q_x - P_y) \, dx \wedge dy. \end{aligned}$$

This is the surface form of $\text{curl } F$. On an oriented surface,

$$\int_S d\omega = \iint_S (\text{curl } F) \cdot n \, dS.$$

Therefore

$$\oint_{\partial S} F \cdot dr = \iint_S (\text{curl } F) \cdot n \, dS.$$

5. Gauss' theorem. Take $\Omega = E$, a solid region, and take the flux form

$$\omega = P \, dy \wedge dz + Q \, dz \wedge dx + R \, dx \wedge dy.$$

Then

$$d\omega = (P_x + Q_y + R_z) \, dx \wedge dy \wedge dz = (\text{div } F) \, dV.$$

The boundary integral of ω is the outward flux:

$$\int_{\partial E} \omega = \iint_{\partial E} F \cdot n \, dS.$$

Therefore

$$\iint_{\partial E} F \cdot n \, dS = \iiint_E \text{div } F \, dV.$$

Why Orientation Matters

Every boundary has a positive direction chosen by the orientation of the inside. Reversing the orientation reverses the sign of the boundary integral. This is why path direction matters for line integrals, normal direction matters for flux, and the right-hand rule matters for Stokes' theorem.

How Conservative Fields Fit In

If a field is conservative, then its “interior derivative” is zero:

$$\text{curl}(\nabla\phi) = 0.$$

Therefore there is no local rotation to accumulate. On a domain with no holes, this forces every closed-loop integral to be zero. If the domain has a hole, a closed curve may surround missing interior, so the local derivative inside the visible domain no longer explains the boundary integral.

8 Step 8: Gravity Field and Its Properties

Gravity is a useful example because it connects all the main ideas of this lecture: potential functions, conservative fields, flux, and Gauss’ formula.

8.1 The Gravity Field of a Point Mass

Let a mass M be located at a point

$$a \in \mathbb{R}^3.$$

The Newtonian gravity field at a point $x \neq a$ is

$$g(x) = -GM \frac{x - a}{\|x - a\|^3}.$$

Here G is the gravitational constant. The vector $g(x)$ is the gravitational acceleration field. If a test mass m is placed at x , then the force on it is

$$\mathbf{F}(x) = m g(x).$$

The minus sign means the field points toward the source mass.

Potential of the Gravity Field

Define the gravitational potential by

$$V(x) = -\frac{GM}{\|x - a\|}.$$

Then

$$g = -\nabla V.$$

So gravity is a conservative field away from the source point.

The field has two local properties away from the mass:

$$\boxed{\text{curl } g = 0, \quad \text{div } g = 0, \quad x \neq a.}$$

The first equation says gravity has no local circulation. The second says empty space has no mass source.

Where Is the Mass?

The equations $\text{curl } g = 0$ and $\text{div } g = 0$ are true only away from the source. The point a is singular. The mass is hidden exactly at the place where the field formula is not defined.

8.2 From the Inverse-Square Law to Gauss' Law

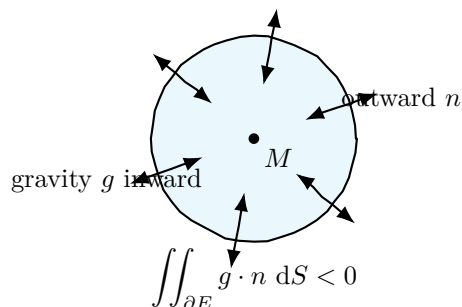
Take a sphere S_r centered at the mass point a , with outward unit normal n . On this sphere,

$$g = -\frac{GM}{r^2}n.$$

Therefore

$$\iint_{S_r} g \cdot n \, dS = \iint_{S_r} \left(-\frac{GM}{r^2} \right) dS = -\frac{GM}{r^2} 4\pi r^2 = \boxed{-4\pi GM}.$$

The radius cancels. This is the geometric signature of the inverse-square law.



For any closed surface ∂E not passing through the source point,

$$\boxed{\iint_{\partial E} g \cdot n \, dS = \begin{cases} -4\pi GM, & a \in E, \\ 0, & a \notin E. \end{cases}}$$

Thus the total outward flux of gravity through a closed surface measures the mass inside, with a negative sign because gravity points inward.

Gauss' Law for Gravity

For a mass density $\rho_m(x)$, the gravity field satisfies

$$\operatorname{div} g = -4\pi G \rho_m.$$

Equivalently, for any region E ,

$$\iint_{\partial E} g \cdot n \, dS = -4\pi G \iiint_E \rho_m \, dV = -4\pi G M_{\text{inside}}.$$

This is Gauss' formula applied to the gravity field.

Since $g = -\nabla V$, Gauss' law can also be written as a Poisson equation for the potential:

$$\operatorname{div} g = \operatorname{div}(-\nabla V) = -\Delta V.$$

So

$$\Delta V = 4\pi G \rho_m.$$

8.3 What Gauss' Formula Simplifies

Suppose the mass distribution is spherically symmetric about the origin. Then symmetry forces the gravity field to have the form

$$g(x) = g_r(r) \hat{r}, \quad r = \|x\|, \quad \hat{r} = \frac{x}{\|x\|}.$$

On the sphere $\|x\| = r$, the quantity $g \cdot n$ is constant and equals $g_r(r)$. Therefore Gauss' law gives

$$4\pi r^2 g_r(r) = -4\pi G M(r),$$

where $M(r)$ is the mass inside the sphere of radius r . Hence

$$g(x) = -\frac{GM(r)}{r^2} \hat{r}.$$

For a thin spherical shell of total mass M and radius R ,

$$M(r) = 0 \quad (r < R), \quad M(r) = M \quad (r > R).$$

Thus

$$g(x) = 0 \quad (r < R),$$

and

$$g(x) = -\frac{GM}{r^2} \hat{r} \quad (r > R).$$

This is the shell theorem: inside a uniform spherical shell the gravity field is zero, and outside the shell it is the same as the field of a point mass placed at the center.

Gauss Versus Stokes for Gravity

Gauss' formula detects the mass source:

$$\iint_{\partial E} g \cdot n \, dS = -4\pi G M_{\text{inside}}.$$

Stokes' formula detects circulation. Since $\text{curl } g = 0$ away from mass sources,

$$\oint_C g \cdot dr = 0$$

whenever C bounds a surface that avoids the singularities of the field. Gravity is a source field, not a rotational field.

9 Exercises

1. Let

$$S_R : x^2 + y^2 + z^2 = R^2$$

be a thin spherical shell centered at the origin, with constant surface mass density σ . A point mass m is located at

$$a = (0, 0, \ell).$$

Compute the gravitational force on the point mass due to the spherical shell:

$$\mathbf{F}(a) = Gm\sigma \iint_{S_R} \frac{r - a}{\|r - a\|^3} \, dS, \quad r = (x, y, z).$$

Show that, if

$$M = 4\pi R^2 \sigma$$

is the total mass of the shell, then

$$\mathbf{F}(a) = \begin{cases} 0, & |\ell| < R, \\ -GmM \frac{a}{\|a\|^3}, & |\ell| > R. \end{cases}$$

Equivalently, since $a = (0, 0, \ell)$,

$$\mathbf{F}(a) = \begin{cases} 0, & |\ell| < R, \\ -4\pi Gm\sigma R^2 \frac{\ell}{|\ell|^3} \mathbf{k}, & |\ell| > R. \end{cases}$$

Think about the two cases

$$|\ell| < R \quad \text{and} \quad |\ell| > R.$$

What happens when $|\ell| = R$?

Hint. By symmetry, the force must point in the z -direction. This is the shell theorem: a uniform spherical shell produces zero gravitational field inside the shell, and outside the shell it acts like all of its mass were concentrated at the center.

Computation. Instead of computing the vector integral directly, first compute the scalar potential integral

$$I(a) = \iint_{S_R} \frac{1}{\|r - a\|} \, dS.$$

Since the sphere is rotationally symmetric, $I(a)$ depends only on

$$\rho = \|a\| = |\ell|.$$

So we may take

$$a = (0, 0, \rho)$$

and use spherical coordinates on the shell:

$$r = (R \sin \phi \cos \theta, R \sin \phi \sin \theta, R \cos \phi),$$

$$0 \leq \theta \leq 2\pi, \quad 0 \leq \phi \leq \pi.$$

Then

$$\|r - a\| = \sqrt{R^2 + \rho^2 - 2R\rho \cos \phi}, \quad dS = R^2 \sin \phi \, d\phi \, d\theta.$$

Thus

$$\begin{aligned} I(\rho) &= \int_0^{2\pi} \int_0^\pi \frac{R^2 \sin \phi}{\sqrt{R^2 + \rho^2 - 2R\rho \cos \phi}} \, d\phi \, d\theta \\ &= 2\pi R^2 \int_0^\pi \frac{\sin \phi}{\sqrt{R^2 + \rho^2 - 2R\rho \cos \phi}} \, d\phi. \end{aligned}$$

Let $u = \cos \phi$. Then

$$\begin{aligned} I(\rho) &= 2\pi R^2 \int_{-1}^1 \frac{du}{\sqrt{R^2 + \rho^2 - 2R\rho u}} \\ &= \frac{2\pi R}{\rho} (R + \rho - |R - \rho|). \end{aligned}$$

Therefore

$$I(\rho) = \begin{cases} 4\pi R, & 0 \leq \rho < R, \\ \frac{4\pi R^2}{\rho}, & \rho > R. \end{cases}$$

Now differentiate with respect to the position a . Since

$$\nabla_a \left(\frac{1}{\|r - a\|} \right) = \frac{r - a}{\|r - a\|^3},$$

we have

$$\iint_{S_R} \frac{r - a}{\|r - a\|^3} dS = \nabla_a I(a).$$

For a radial function $I(a) = I(\rho)$,

$$\nabla_a I(a) = I'(\rho) \frac{a}{\rho}.$$

Hence

$$\nabla_a I(a) = \begin{cases} 0, & \rho < R, \\ -4\pi R^2 \frac{a}{\|a\|^3}, & \rho > R. \end{cases}$$

Multiplying by $Gm\sigma$ gives

$$\mathbf{F}(a) = \begin{cases} 0, & |\ell| < R, \\ -GmM \frac{a}{\|a\|^3}, & |\ell| > R, \end{cases} \quad M = 4\pi R^2 \sigma.$$

When $|\ell| = R$, the point mass lies on the shell, so the force integral has a singularity on the surface and is not defined as an ordinary improper integral.

Remark on the boundary case. For an ideal infinitely thin shell, the physically meaningful quantities are the one-sided limits. If n is the outward unit normal at the point $a \in S_R$, then

$$\mathbf{F}_{\text{inside}} = 0, \quad \mathbf{F}_{\text{outside}} = -\frac{GmM}{R^2} n.$$

The Cauchy principal value is the average of these two limits:

$$\text{p. v. } \mathbf{F}(a) = \frac{1}{2} (\mathbf{F}_{\text{inside}} + \mathbf{F}_{\text{outside}}) = -\frac{GmM}{2R^2} n.$$

Equivalently, since $M = 4\pi R^2 \sigma$,

$$\text{p. v. } \mathbf{F}(a) = -2\pi Gm\sigma n.$$

This principal value is a symmetric mathematical prescription. A real shell has finite thickness or microscopic structure, so the actual force at the material surface depends on the physical model used near the point.

10 Summary

- A scalar line integral uses ds and accumulates density along length.
- A vector line integral uses $F \cdot dr$ and accumulates tangential component; it changes sign when orientation is reversed.

- A conservative vector field has the form $F = \nabla\phi$.

- For conservative fields,

$$\int_C F \cdot dr = \phi(B) - \phi(A).$$

- On a simply connected plane region, $Q_x - P_y = 0$ is the test for a conservative field.
- A parametrized surface $X(u, v)$ has area element

$$dS = \|X_u \times X_v\| du dv.$$

- A scalar surface integral uses

$$\iint_S f dS.$$

- A vector surface integral, or flux integral, uses

$$\iint_S F \cdot n dS.$$

- Scalar surface integrals do not depend on orientation; flux integrals do.
- For a graph $z = g(x, y)$, upward vector area is

$$(-g_x, -g_y, 1) dx dy.$$

- The divergence of $F = (P, Q, R)$ is

$$\operatorname{div} F = P_x + Q_y + R_z.$$

- Gauss' formula, or the divergence theorem, says

$$\iint_{\partial E} F \cdot n dS = \iiint_E \operatorname{div} F dV,$$

where ∂E is closed and outward oriented.

- Stokes' formula says

$$\oint_{\partial S} F \cdot dr = \iint_S (\operatorname{curl} F) \cdot n dS.$$

The orientation of ∂S is induced by the chosen normal using the right-hand rule.

- The unifying principle is

$$\int_{\partial\Omega} \omega = \int_{\Omega} d\omega :$$

boundary integral equals accumulated interior derivative.

- The Newtonian gravity field of a point mass is

$$g(x) = -GM \frac{x - a}{\|x - a\|^3} = -\nabla \left(-\frac{GM}{\|x - a\|} \right),$$

so it is conservative away from the mass source.

- Gauss' law for gravity says

$$\operatorname{div} g = -4\pi G \rho_{\text{m}}, \quad \iint_{\partial E} g \cdot n \, dS = -4\pi G M_{\text{inside}}.$$

Mass appears as divergence, while $\operatorname{curl} g = 0$ away from sources.